ANALYSIS REPORT

Building a Text Statistics Analyzer using Python Fundamentals

The objective of this project was to develop a Text Statistics Analyzer using Python fundamentals and core libraries such as **Regular Expressions, NumPy, Pandas, and Matplotlib**. The system processes a complete collection of text documents and performs text exploration, pattern extraction, statistical analysis, and visualization.

The **BBC News dataset** was selected for this project. It contains **2,225 news articles** distributed across five categories: Business, Entertainment, Politics, Sport, and Technology. The complete dataset was analysed rather than using a small sample.

The project was implemented in a Jupyter Notebook and produced document-level statistics, extracted patterns, statistical summaries, multiple visualizations, Pandas-based word analysis, and processed text outputs.

Dataset and Text Processing

Each of the 2,225 BBC news articles was loaded using Python file I/O. The text was processed using regular expressions to identify words and sentences.

For every document, the analysis recorded:

- Document ID
- Filename
- Category
- Word count
- Character count
- Sentence count
- Average word length
- Average sentence length

The document statistics were stored in a Pandas DataFrame and exported as document_statistics.csv.

Overall Dataset Statistics

Metric	Result
Total Documents	2,225
Total Characters	5,039,982
Total Words	855,598
Total Sentences	44,468
Average Word Length	4.66 characters
Average Sentence Length	19.24 words

The results show that the dataset contains a large volume of news text, making it suitable for demonstrating fundamental NLP preprocessing and statistical techniques.

3. Word Frequency Analysis

A word-frequency dictionary was created from the complete corpus and converted into a NumPy array for statistical analysis.

The frequency visualization showed that a relatively small number of words occur extremely frequently, while many other words occur much less frequently.

The **Top 20 Words visualization** clearly showed the dominance of common English function words such as:

- the
- to
- of
- and
- a
- in
- for
- is
- that

These words occur frequently because they are grammatical words that appear across almost every category of news article.

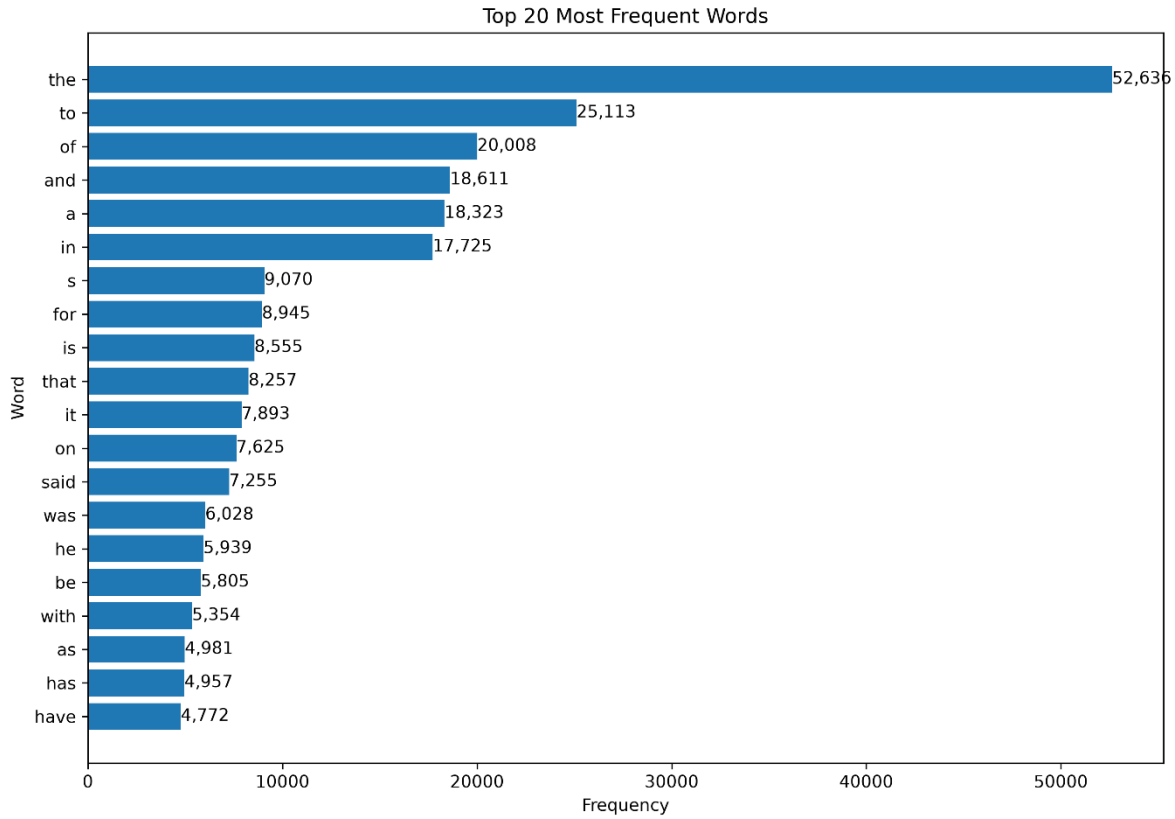


Figure 1: Top 20 Most Frequent Words

The horizontal bar chart makes the difference between high-frequency and lower-frequency words easy to observe. Value labels were added to the bars so that the frequencies can be directly compared.

One interesting preprocessing observation was the occurrence of **"s"** as a separate token. This can occur when contractions or possessive forms are separated during regex-based word extraction. This demonstrates one of the limitations of simple rule-based tokenization.

4. Word Length Distribution

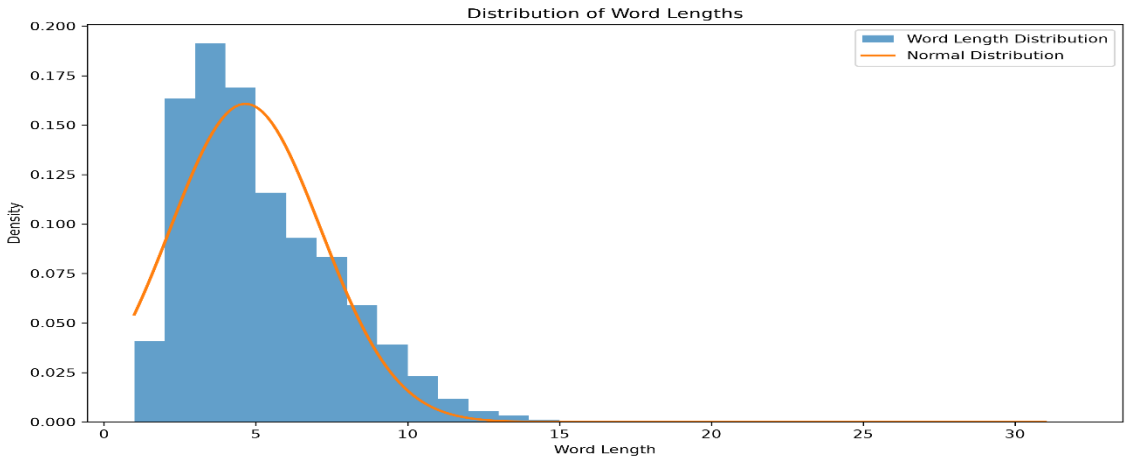


Figure 2: Word Length Distribution

The word-length analysis examined the distribution of the number of characters in individual words.

The histogram shows that shorter and medium-length words occur much more frequently than very long words. A normal distribution curve was also plotted over the histogram to provide a reference for the overall distribution.

The analysis also identified words longer than 10 characters using the Pandas-based filtering operation. These longer words are relatively uncommon compared with shorter words.

This result is expected in English news text because common grammatical words and frequently used content words tend to be relatively short.

5. Character-Level Analysis

Character frequency analysis was performed to understand the structure of the corpus at the character level.

The most frequently occurring characters were mainly common English letters, with **e being the most frequent character**.

The analysis also calculated the number of vowels and consonants:

- Vowels: approximately **1.51 million**
- Consonants: approximately **2.48 million**
- Vowel-to-consonant ratio: approximately **0.61**

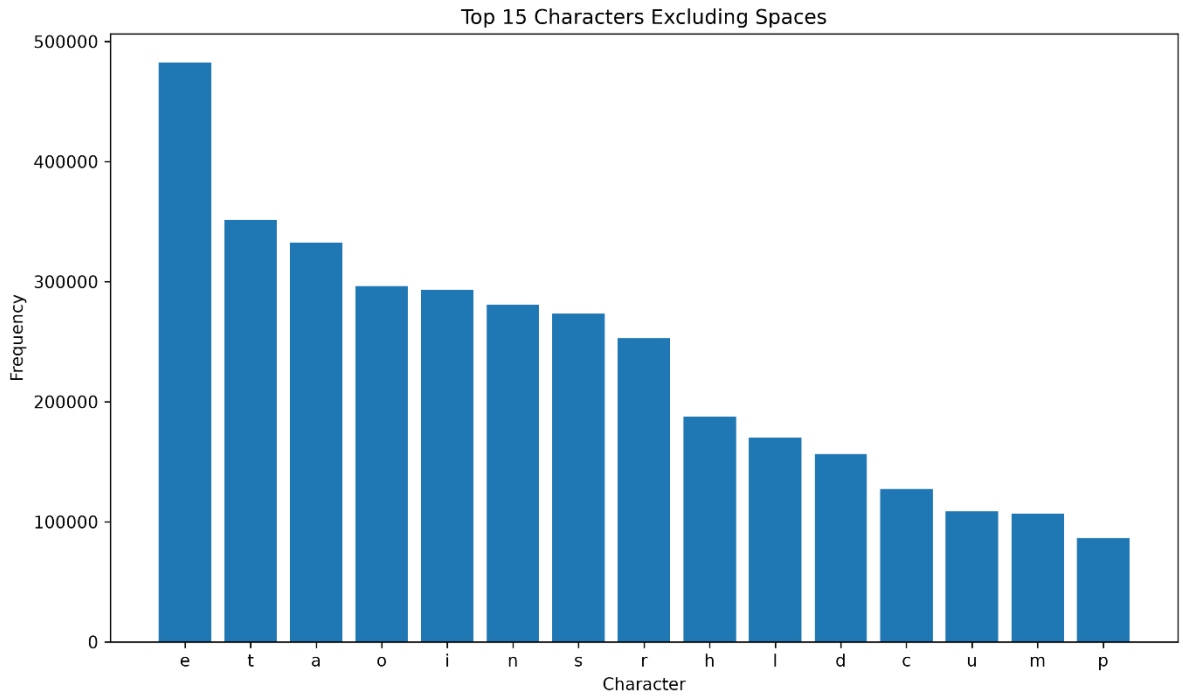


Figure 3: Top 15 Characters

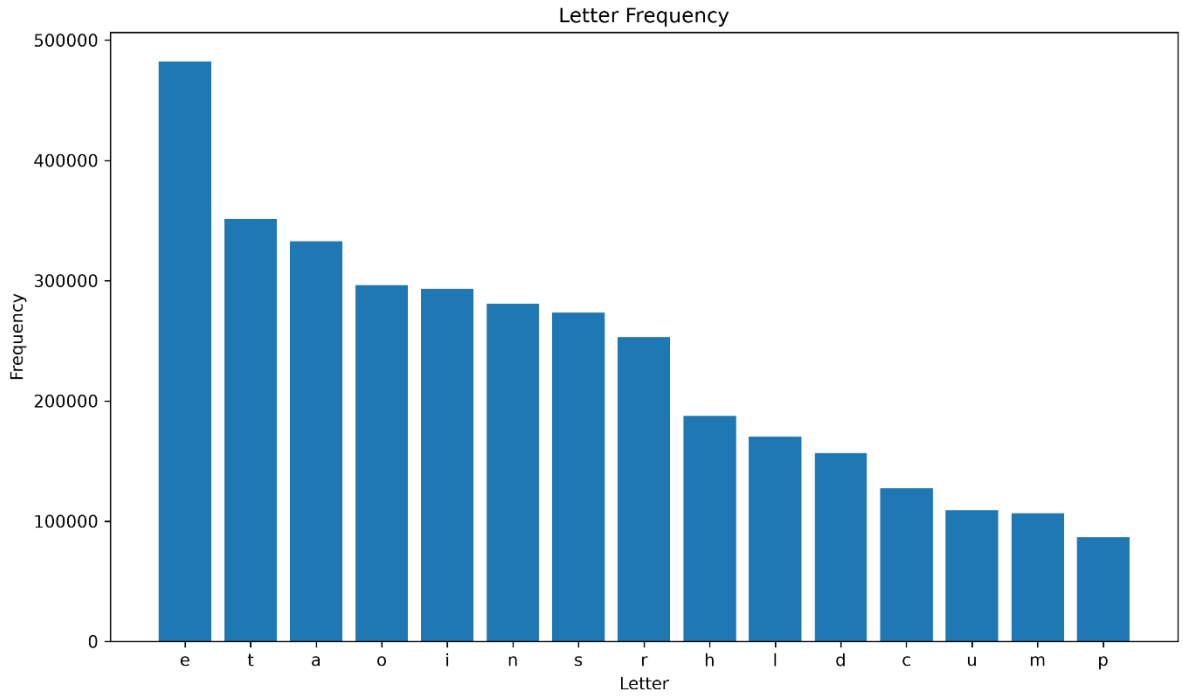


Figure 4: Letter Frequency

6. Sentence Length Analysis

Sentences were extracted using regular expressions and their lengths were calculated based on the number of words in each sentence.

The analysis produced:

- **44,468 sentences**
- Average sentence length: **19.24 words**

Figure 6: Sentence Length Box Plot

[Insert the sentence_length_boxplot.png visualization here]

The box plot shows the central distribution of sentence lengths as well as unusually long sentences.

Some sentences were significantly longer than the typical sentence. These outliers can occur because news articles sometimes contain long sentences with multiple clauses, quotations, lists, or several pieces of information.

The sentence-length visualization therefore provides useful information about the writing structure of the BBC news corpus.

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7. Regular Expression Pattern Extraction

Regular expressions were used to search the complete dataset for emails, phone numbers, hashtags, mentions, URLs, dates, and currency values.

Pattern	Occurrences	Unique Values
Emails	0	0
Phone Numbers	0	0
Hashtags	30	6
Mentions	0	0
URLs	3	2
Dates	2	1
Currency Values	1,543	550

The most significant pattern was the occurrence of **currency values**, with 1,543 occurrences. This is reasonable because the BBC dataset contains business and financial news.

Six unique hashtags were found, while no email addresses, phone numbers, or mentions were detected using the implemented patterns.

Two unique URL domains were identified:

- dec.org.uk
- arsenal.com

The extracted patterns were stored in extracted_patterns.json.

One of the main challenges in this stage was creating regular expressions that could identify the required patterns without producing excessive false matches. Different formats also required separate regex patterns.

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8. Visualization and Category Analysis

A **2×2 dashboard** was created to combine important statistics from the text analysis into a single visualization.

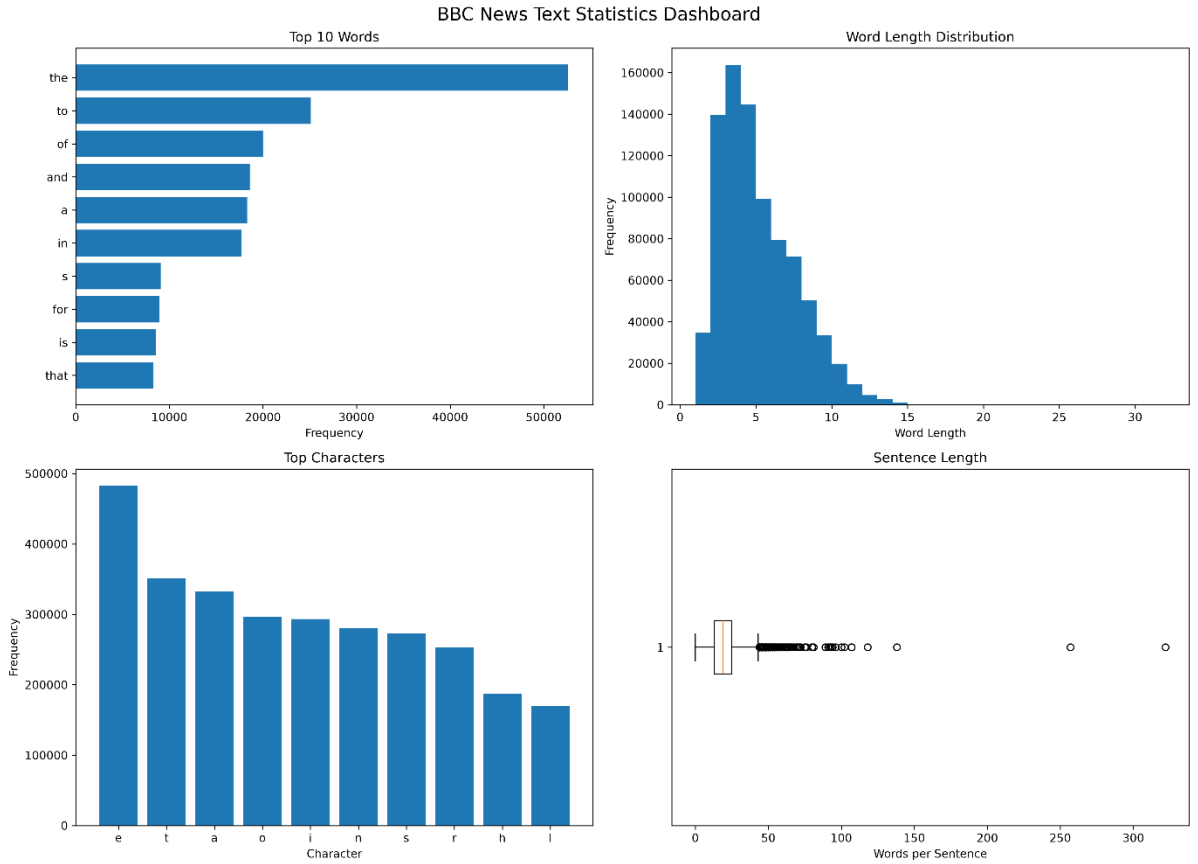


Figure 7: Text Statistics Dashboard

The dashboard provides a compact overview of the major findings, including word frequency, word-length distribution, character frequency, and sentence statistics.

A custom visualization was also created to compare the average number of words per document across the five BBC categories.

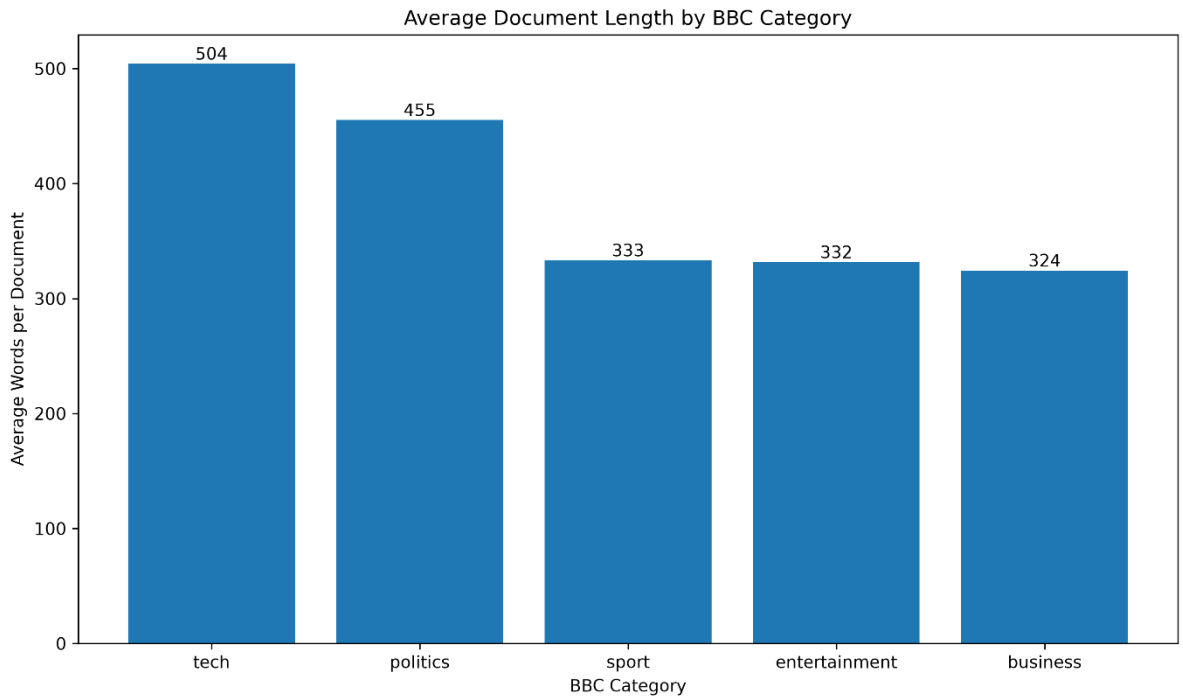


Figure 8: Average Document Length by Category

The approximate average document lengths were:

Category	Average Words per Document
Technology	504
Politics	455
Sport	333
Entertainment	332
Business	324

This visualization shows that the categories do not have identical document lengths. Technology and politics articles had higher average word counts, while business and entertainment articles had lower average word counts in this dataset.

9. Zipf-Like Frequency Pattern

The word-frequency results show a pattern where a small number of words occur very frequently while a large number of words occur relatively rarely.

This is consistent with the general behavior described by **Zipf's law**, where word frequency tends to decrease rapidly as word rank increases.

The project did not perform a formal mathematical fit of Zipf's law, so the observation is based on the frequency distribution rather than a calculated Zipf exponent.

This pattern is clearly visible in the **Top 20 Words visualization**, where the most frequent words have considerably higher frequencies than words further down the ranking.

10. Vocabulary and Pandas Analysis

A Pandas DataFrame was created with every unique word as the index and the following attributes:

- Frequency
- Word length
- First character
- Vowel count

Group-by operations were performed using word length and first character.

The analysis also generated:

- Top 50 words by frequency
- Words longer than 10 characters
- Word-length pivot table
- Word-length cross-tabulation
- First-character statistics

This analysis demonstrated how text can be converted into structured tabular data and then analyzed using standard Pandas operations.

The large difference between highly frequent words and less frequent vocabulary also indicates that the corpus contains repeated common vocabulary alongside a much larger collection of less frequent words.

11. Correlations and Interesting Observations

The document-level statistics showed a strong relationship between document word count and character count. This is expected because documents containing more words generally contain more characters.

The category analysis also showed noticeable differences in document length.

Another important observation was that the most frequent words were mostly common grammatical words rather than specialized topic words. Therefore, raw word frequency alone does not necessarily identify the most meaningful terms in a document collection.

The character-level analysis similarly showed that a small number of common English characters dominate the corpus.

12. Challenges Faced and Solutions

Dataset Handling

The dataset contained a README file in addition to the news articles. The processing logic was restricted to the five BBC category folders so that exactly **2,225 news documents** were analyzed.

Regex Tokenization

Simple regular expressions can create unwanted tokens when processing contractions and possessive forms. The appearance of tokens such as "s" demonstrated this limitation.

Sentence Extraction

Sentence splitting using punctuation-based regex is not perfect because real news articles contain quotations, abbreviations, lists, and complex sentence structures. These cases can result in unusually long sentence measurements.

Pattern Extraction

Emails, phone numbers, URLs, dates, hashtags, and currency values all have different structures. Separate regex patterns were therefore required for each pattern type.

Visualization

The dataset contains thousands of unique words and characters, so displaying every value would make the charts difficult to interpret. The visualizations therefore focus on the most frequent values and important distributions.

13. Conclusion

The Text Statistics Analyzer successfully processed the complete **2,225-document BBC News dataset** using Python fundamentals, Regular Expressions, NumPy, Pandas, and Matplotlib.

The analysis produced **5,039,982 characters, 855,598 words, and 44,468 sentences**, with an average word length of approximately **4.66 characters** and an average sentence length of approximately **19.24 words**.

The project demonstrated that basic Python techniques can be used to perform meaningful text analysis without relying on specialized NLP libraries. Word-frequency analysis revealed a strong concentration of common words, while character-level analysis showed the dominance of common English letters. Sentence analysis identified both typical sentence lengths and unusually long sentences.

Regular expression analysis successfully extracted hashtags, URLs, dates, and currency values while also demonstrating the challenges associated with extracting structured patterns from unrestricted text.

The visualizations provided a clear representation of word frequencies, word lengths, character distributions, sentence lengths, and differences between news categories. Overall, the project establishes a strong foundation for progressing from basic text processing toward more advanced NLP techniques such as n-gram analysis, text classification, embeddings, and machine learning.

visualization insights.